



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering

Academic Year 2025 – 2026 (Even Semester)

Degree, Semester & Branch: II Semester B.E. CSE 'B'

Course Code & Title: CS25C07 Object Oriented Programming

Name of the Faculty member (s): Dr.B.Vijayalakshmi, ASP-II/CSE

Innovative Practice Description

- **Unit / Topic: Module 2 / Class, Objects, Constructors**
- **Course Outcome: CO2**
- **Activity Chosen: Problem Based Learning**
- **Justification:**

Classes, objects, and constructors form the foundation of Object-Oriented Programming. Students often understand the syntax of classes and objects but face challenges in applying these concepts to solve real-world programming problems. Problem Based Learning (PBL) enables students to learn by analyzing practical scenarios, identifying requirements, and designing appropriate classes and objects. This approach promotes active participation, problem-solving skills, critical thinking, and collaborative learning. By solving realistic problems, students gain a deeper understanding of object creation, initialization through constructors, and object-oriented design principles.
- **Time Allotted for the Activity: 15 Minutes**
- **Details of the Implementation:**
 - The concepts of classes, objects, constructors, destructors, and static members were explained with suitable examples and C++ programs.
 - After the completion of the topic, a Problem Based Learning (PBL) activity was conducted to enable students to apply object-oriented concepts in real-world scenarios.
 - Students were divided into five groups consisting of 4–5 members each.
 - Each group was assigned one of the following problem statements:

Group 1: Student Information System

 - Develop a class Student to store Student ID, Name, Department, and CGPA. Use a constructor to initialize student details and display the information of multiple students.

Group 2: Bank Account Management System

 - Design a class BankAccount to maintain Account Number, Account Holder Name, and Balance. Use constructors to initialize account details and create functions for deposit and withdrawal operations.

Group 3: Library Book Management System

- Create a class Book to store Book ID, Title, Author Name, and Price. Use constructors for initializing book information and display the details of available books.

Group 4: Employee Payroll System

- Develop a class Employee containing Employee ID, Employee Name, Designation, and Salary. Use constructors to initialize employee data and display payroll information.

Group 5: Mobile Phone Inventory System

- Create a class MobilePhone with attributes such as Brand, Model, RAM, Storage, and Price. Initialize the details using constructors and display product information.
- Students were instructed to analyze their assigned problem and identify:
 - Appropriate class name.
 - Data members required.
 - Suitable constructor(s) for object initialization.
 - Member functions required for processing and displaying data.
 - Objects needed to represent real-world entities.
- Each group discussed the problem requirements and prepared a class design diagram as shown in Fig.1.
- Students developed the class structure and demonstrated how constructors can be used to initialize object attributes efficiently.
- The groups presented their proposed solutions to the class and explained how object-oriented concepts were applied in solving the assigned problem.
- During the presentation, students justified their choice of data members, constructors, and member functions.
- The faculty member facilitated discussions, compared different solutions, highlighted best programming practices, and clarified doubts regarding class design, object creation, and constructor usage.
- Finally, students reflected on how object-oriented concepts can be applied to model various real-world systems effectively.

• CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO2	3	3	2	-	1	-	-	1	1	-	1	3	3	1

(1 – Low 2 – Moderate 3 – High)

• PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1
	3	3	3	1	1	3
Justification for correlation		Students analyze the requirements for the		Students work collaboratively in groups to discuss and	Students communicate their design decisions and	Students apply programming knowledge to

	Students apply object-oriented programming fundamentals to design classes and objects for a given problem.	problem and identify suitable class members and constructors.	Students develop a solution by designing appropriate classes, objects, and member functions.	formulate solutions.	implementation ideas during presentations.	solve practical software development problems.
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Images / Screenshot of the practice:

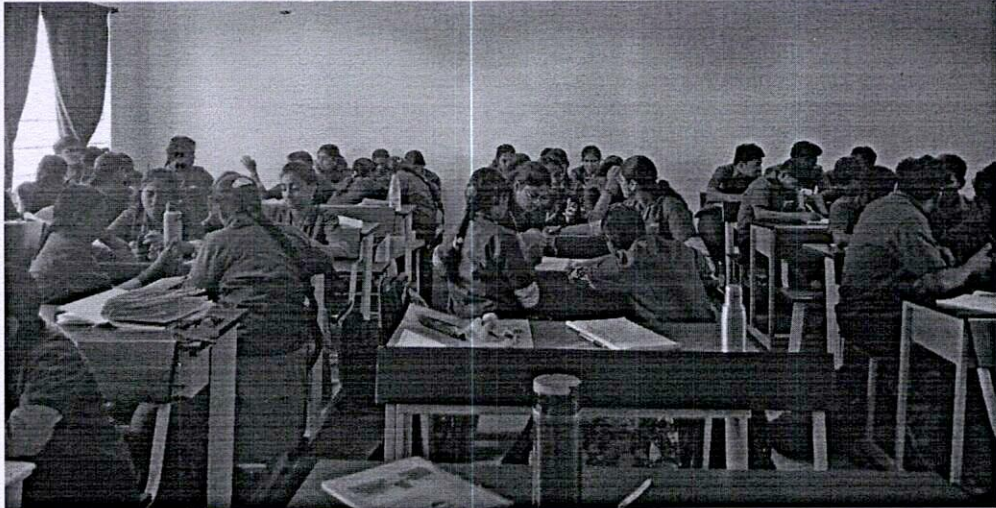


Fig 1. Students discussing the problem requirements and identifying class components.

Reflective Critique:

❖ **Feedback of practice from students and other stakeholders:**

- Students stated that solving a real-world problem helped them understand the practical use of classes and constructors.
- Students appreciated the group discussion as it enabled them to learn different approaches to designing class structures.
- Students reported improved confidence in creating object-oriented solutions.

❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- Enhanced understanding of classes, objects, and constructors through practical application.
- Improved problem-solving and analytical thinking skills.
- Encouraged teamwork and collaborative learning.
- Increased student engagement compared to traditional lecture-based instruction.

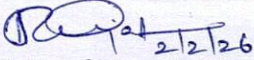
❖ **Challenges faced in implementation:**

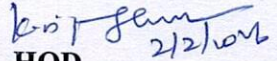
- Some students initially found it difficult to identify suitable data members and member functions.

- A few groups required guidance in selecting appropriate constructor types.
- Time management was needed to ensure all groups could present their solutions.

References:

- <https://www.teachthought.com/learning/problem-based-learning/>
- https://www.ritrjpm.ac.in/images/computer-science/2022-2023/1_GS_CS8592_problem_based_learning.pdf


Signature of Faculty Member


HOD